

CoolGaN™ G5

CoolGaN™ Transistor 650 V G5

Infineon's CoolGaN™ is a highly efficient gallium nitride (GaN) transistor designed for power conversion at 650 V. It enables higher power density, supports reduced system BOM cost, and facilitates miniaturized form factors. Produced using 200 mm (8 inch) wafer technology and fully automated production lines, it features narrow production tolerances and the highest product quality. This makes it suitable for a wide range of applications, from consumer electronics to industrial applications.

Features

- Enhancement mode transistor
- Ultra-fast switching
- No reverse-recovery charge
- Capable of reverse conduction
- Low gate and output charge
- Superior commutation ruggedness
- 2 kV HBM ESD standards

Benefits

- Normally OFF transistor technology ensures safe operation
- Enables rapid and precise power delivery control
- Improves system efficiency and reliability
- Ensures robust performance under challenging conditions

These features collectively make CoolGaN™ a game-changer in the realm of power conversion, offering a compelling combination of efficiency, compactness, and reliability.

Potential applications

Industrial, telecom, datacenter SMPS based on half-bridge hard and soft switching topologies such as totem pole PFC and high frequency LLC, as well as charger and adapter.

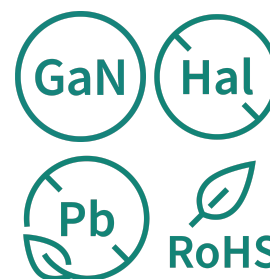
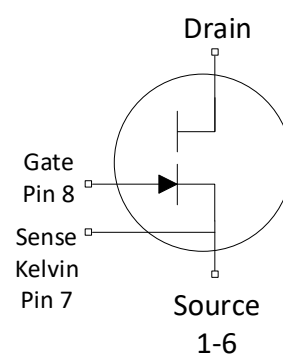
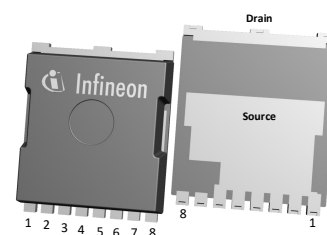
Product validation

Qualified according to relevant JEDEC tests.

Table 1 Key performance parameters

Parameter	Value	Unit
$V_{DS,max}$	650	V
$V_{DS,trans-max}$	900	V
$R_{DS(on),max}$	170	mΩ
$Q_{g,typ}$	1.8	nC
$I_{D,pulse}$	23	A
$Q_{oss} @ 400 V$	14	nC
Q_{rr}	0	nC

TOLL



Part number	Package	Marking	Related links
IGT65R140D2	PG-HSOF-8	65R140D2	see Appendix A



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1 Maximum ratings

at $T_j = 25^\circ\text{C}$, unless otherwise specified. Stresses beyond max ratings may cause permanent damage to the device. For optimum lifetime and reliability, Infineon recommends operating conditions that do not continuously exceed 80% of the maximum ratings stated (unless otherwise explicitly stated). For further information, contact your local Infineon sales office.

Table 2 Maximum ratings

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Drain source voltage, continuous	$V_{DS,max}$	-	-	650	V	$V_{GS} = 0\text{ V}$; derating recommendation according JEDEC JEP198
Leakage current at drain source transient voltage	$I_{DS,trans}$	-	-	4.8	mA	$V_{GS} = 0\text{ V}$; $V_{DS,trans} = 900\text{ V}$
Drain source voltage transient	$V_{DS,trans}$	-	-	900	V	<1% duty cycle; <1 μs ; 1 million pulses
Drain source voltage, pulsed	$V_{DS,pulsed}$	-	-	750	V	$T_j = 25^\circ\text{C}$; $V_{GS} \leq 0\text{ V}$; cumulated stress time $\leq 10\text{ h}$
						$T_j = 125^\circ\text{C}$; $V_{GS} \leq 0\text{ V}$; cumulated stress time $\leq 1\text{ h}$
Switching surge voltage, pulsed	$V_{DS,surge}$	-	-	750	V	DC bus voltage = 700 V; turn off $V_{DS,pulse} = 750\text{ V}$; turn on $I_{D,pulse} = 10\text{ A}$; $f \leq 100\text{ kHz}$; $T_j = 105^\circ\text{C}$; $t \leq 100\text{ s}$ (10 million pulses)
Continuous current, drain source ¹⁾	I_D	-	-	13	A	$T_C = 25^\circ\text{C}$; $T_j = T_{j,max}$
Pulsed current, drain source ²⁾	$I_{D,pulse}$	-23	-	23	A	$T_j = 25^\circ\text{C}$; $I_G = 10\text{ mA}$; See Diagram 3, 5
		-14		14		$T_j = 125^\circ\text{C}$; $I_G = 10\text{ mA}$; See Diagram 4, 6
Gate current, continuous ³⁾	$I_{G,avg}$	-	4.6	7.7	mA	$T_j = -55^\circ\text{C}$ to $T_j = 150^\circ\text{C}$; See Table 9
Gate current, pulsed ³⁾	$I_{G,pulsed}$	-0.77	-	0.77	A	$T_j = -55^\circ\text{C}$ to $T_j = 150^\circ\text{C}$; $t_{pulse} = 50\text{ ns}$; $f = 100\text{ kHz}$; See Table 9
Gate source voltage, continuous ³⁾	V_{GS}	-10	-	-	V	$T_j = -55^\circ\text{C}$ to $T_j = 150^\circ\text{C}$; See Diagram 12
Gate source voltage, pulsed ³⁾	$V_{GS,pulse}$	-25	-	-	V	$T_j = -55^\circ\text{C}$ to $T_j = 150^\circ\text{C}$; $t_{pulse} = 50\text{ ns}$; $f = 100\text{ kHz}$; open drain
Power dissipation	P_{tot}	-	-	47	W	$T_C = 25^\circ\text{C}$
Operating junction temperature	T_j	-55	-	150	$^\circ\text{C}$	-
Storage temperature	T_{stg}	-55	-	150	$^\circ\text{C}$	Max shelf life depends on storage conditions
Drain-source voltage slew-rate	dv/dt	-	-	200	V/ns	-

- 1) Limited by $T_{j,max}$. Maximum duty cycle $D = 0.75$
- 2) Defined by design. Not subject to production test.
- 3) Consider the influence of the gate current (I_G) on the drain current (I_D). For further details, see Table 10 (related links) and refer to the Gate Drive Application Note and the Reliability White Paper, or contact your local Infineon sales office.

2 Thermal characteristics

Table 3 Thermal characteristics

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Thermal resistance, junction - case	R_{thJC}	-	-	2.7	°C/W	-
Thermal resistance, junction - ambient	R_{thJA}	-	-	86	°C/W	Device on PCB, minimum footprint
Thermal resistance, junction - ambient for SMD version	R_{thJA}	-	-	62	°C/W	Device on 40 mm*40 mm*1.5 mm epoxy PCB FR4 with 6 cm ² (one layer, 70 µm thickness) copper area for tab (source) connection and cooling. PCB is vertical without air stream cooling.
Reflow soldering temperature	T_{sold}	-	-	260	°C	MSL1

3 Electrical characteristics

at $T_j=25^\circ\text{C}$, unless specified otherwise

Table 4 Static characteristics

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Gate threshold voltage	$V_{GS(th)}$	0.9	1.2	1.6	V	$I_{DS}=1\text{ mA}$; $V_{DS}=10\text{ V}$; $T_j=25^\circ\text{C}$
		-	1	-		$I_{DS}=1\text{ mA}$; $V_{DS}=10\text{ V}$; $T_j=150^\circ\text{C}$
Gate-Source reverse clamping voltage	$V_{GS, clamp}$	-	-	-8	V	$I_{GS}=-1\text{ mA}$
Drain-Source leakage current	I_{DSS}	-	0.39	39	μA	$V_{DS}=650\text{ V}$; $V_{GS}=0\text{ V}$; $T_j=25^\circ\text{C}$
			7.8	-		$V_{DS}=650\text{ V}$; $V_{GS}=0\text{ V}$; $T_j=150^\circ\text{C}$
Drain-Source on-state resistance	$R_{DS(on)}$	-	0.140	0.170	Ω	$I_G=10\text{ mA}$; $I_D=3.1\text{ A}$; $T_j=25^\circ\text{C}$
			0.300	-		$I_G=10\text{ mA}$; $I_D=3.1\text{ A}$; $T_j=150^\circ\text{C}$
Gate resistance ⁴⁾	$R_{G,int}$	-	0.92	-	Ω	LCR impedance measurement; $f=f_{res}$, open drain

⁴⁾ Defined by design. Not subject to production test.

Table 5 Dynamic characteristics

Defined by design. Not subject to production test.

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Input capacitance	C_{iss}	-	130	-	pF	$V_{GS}=0\text{ V}$; $V_{DS}=400\text{ V}$; $f=1\text{ MHz}$
Output capacitance	C_{oss}		22			
Reverse transfer capacitance	C_{rss}		0.31			
Effective output capacitance, time related ⁵⁾	$C_{o(tr)}$	-	35	-	pF	$V_{GS}=0\text{ V}$; $V_{DS}=0\text{ to }400\text{ V}$; $I_D=\text{const}$
Effective output capacitance, energy related ⁶⁾	$C_{o(er)}$	-	26	-	pF	$V_{DS}=0\text{ to }400\text{ V}$
Output charge	Q_{oss}		14	-	nC	
Coss stored energy	E_{oss}		2.1	-	μJ	
Turn-on delay time	$t_{d(on)}$	-	7	-	ns	$I_D=3.1\text{ A}$; $R_{ON}=12\text{ }\Omega$; $R_{OFF}=12\text{ }\Omega$; $R_{SS}=820\text{ }\Omega$; $C_C=1.2\text{ nF}$; $V_{DRV}=12\text{ V}$; see Table 8
Turn-off delay time	$t_{d(off)}$		10			
Rise time	t_r		7			
Fall time	t_f		23			

⁵⁾ $C_{o(tr)}$ is a fixed capacitance that gives the same charging time as C_{oss} while V_{DS} is rising from 0 to 400 V

⁶⁾ $C_{o(er)}$ is a fixed capacitance that gives the same stored energy as C_{oss} while V_{DS} is rising from 0 to 400 V

Table 6 Gate charge characteristics

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Gate charge ⁷⁾	Q_G	-	1.8	-	nC	$V_{GS}=0$ to 3 V; $V_{DS}=400$ V; $I_D=3.1$ A

⁷⁾ Defined by design. Not subject to production test.

Table 7 Reverse conduction characteristics

Parameter	Symbol	Values			Unit	Note / Test condition
		Min.	Typ.	Max.		
Source-Drain reverse voltage	V_{SD}	-	2.0	2.4	V	$V_{GS}=0$ V; $I_{SD}=3.1$ A
Pulsed current, reverse	$I_{SD,pulse}$	-	-	23	A	$I_G=10$ mA
Reverse recovery charge ⁸⁾	Q_{rr}	-	0	-	nC	$I_{SD}=3.1$ A; $V_{DS}=400$ V

⁸⁾ Defined by design. Not subject to production test. Excluding Q_{oss} .

4 Electrical characteristics diagrams

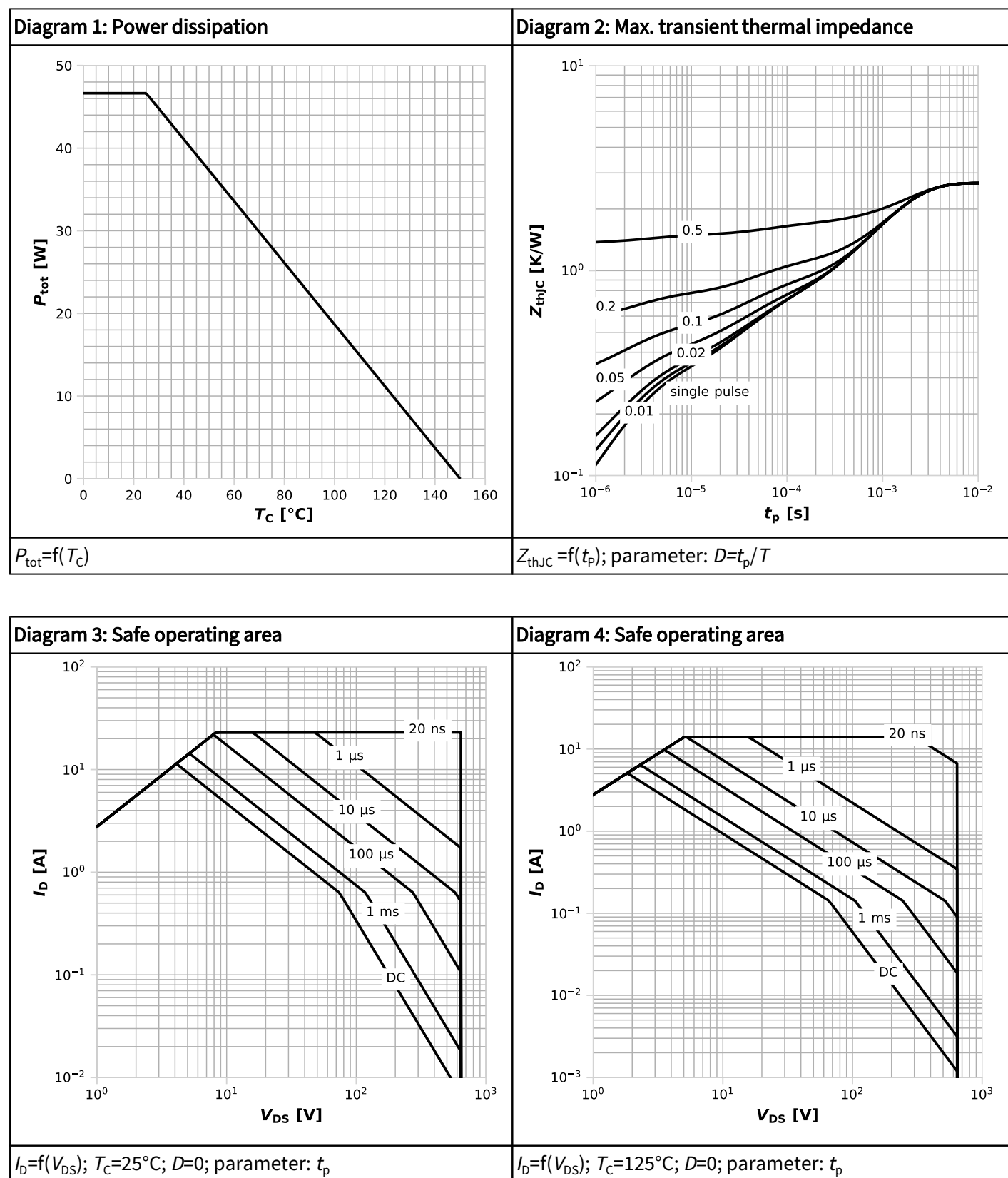
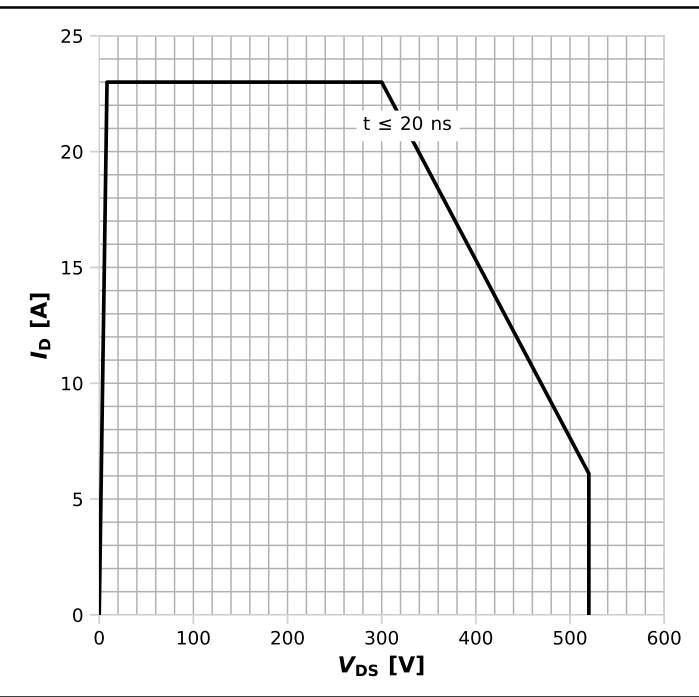
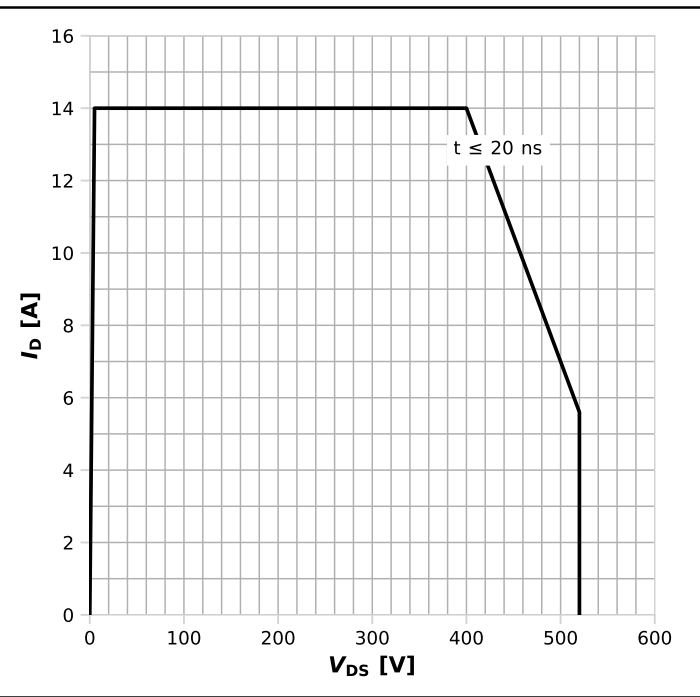


Diagram 5: Repetitive safe operating area



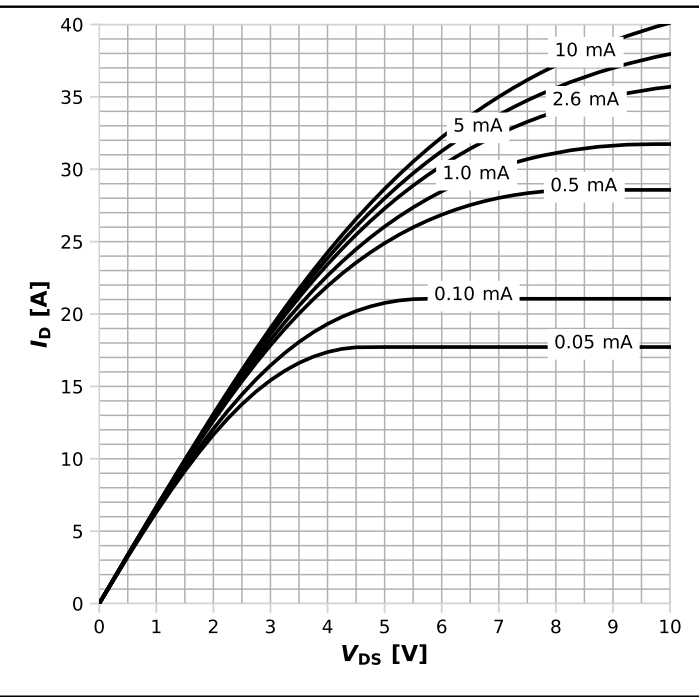
$I_D=f(V_{DS})$; $T_c=25^\circ\text{C}$; $T_j \leq 150^\circ\text{C}$; parameter: t_p

Diagram 6: Repetitive safe operating area



$I_D=f(V_{DS})$; $T_c=125^\circ\text{C}$; $T_j \leq 150^\circ\text{C}$; parameter: t_p

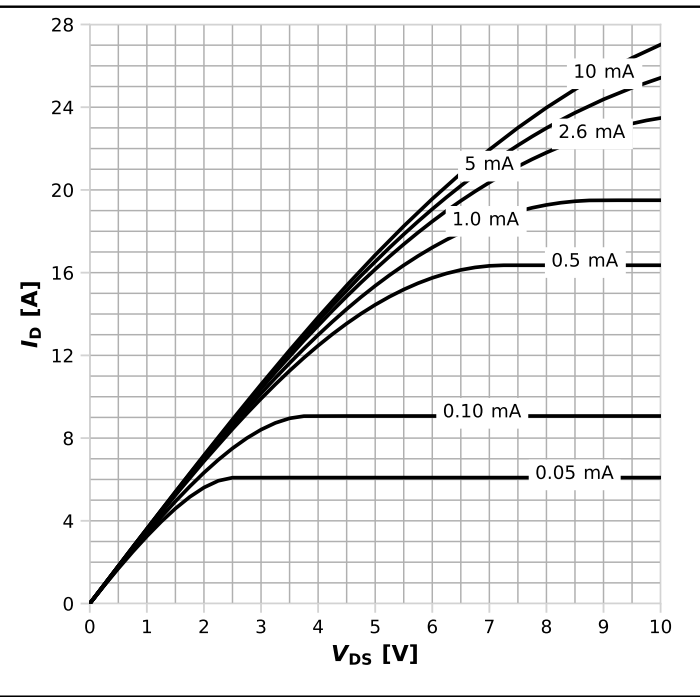
Diagram 7: Typ. static* output characteristics



$I_D=f(V_{DS})$; $T_j=25^\circ\text{C}$; parameter: I_{GS}

* Refer to gate drive application note (see table 10)

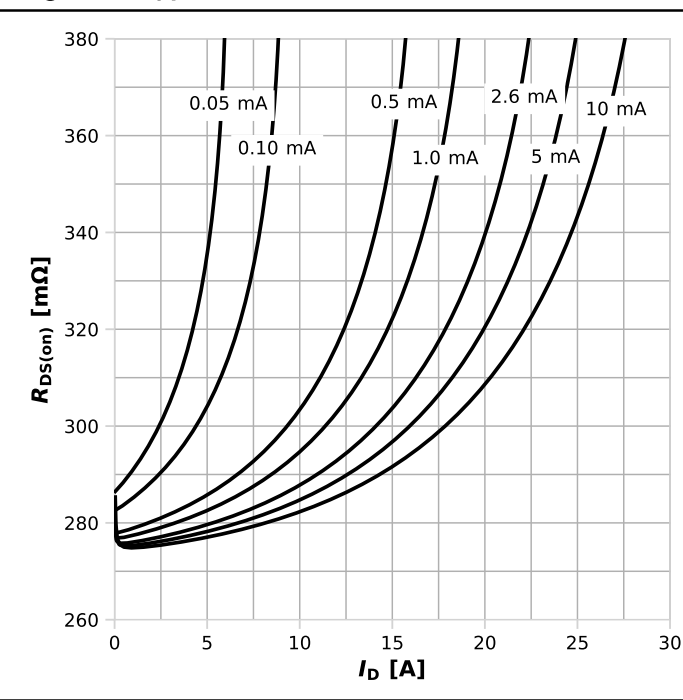
Diagram 8: Typ. static* output characteristics



$I_D=f(V_{DS})$; $T_j=125^\circ\text{C}$; parameter: I_{GS}

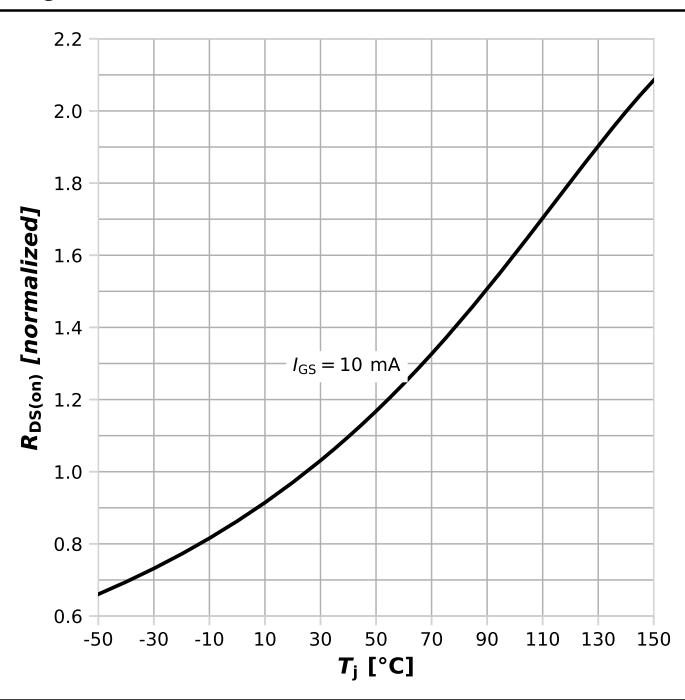
* Refer to gate drive application note (see table 10)

Diagram 9: Typ. Drain-source on-state resistance



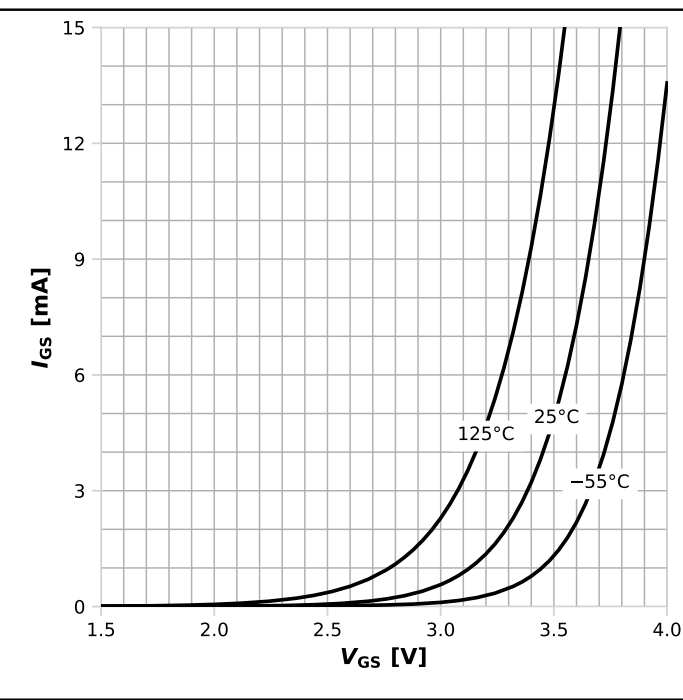
$R_{DS(on)} = f(I_D); T_j = 125^\circ\text{C}; \text{parameter: } I_{GS}$

Diagram 10: Drain-source on-state resistance



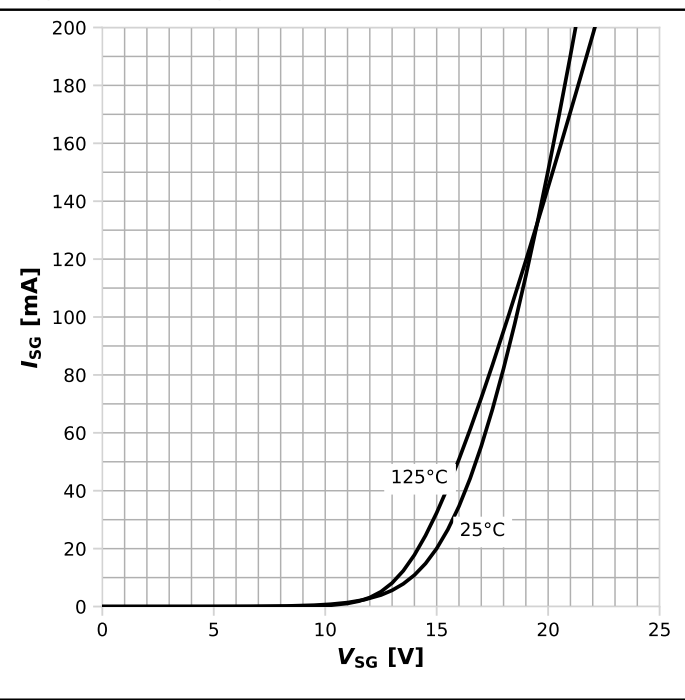
$R_{DS(on)} = f(T_j); I_D = 3.1 \text{ A}$

Diagram 11: Typ. gate characteristics forward



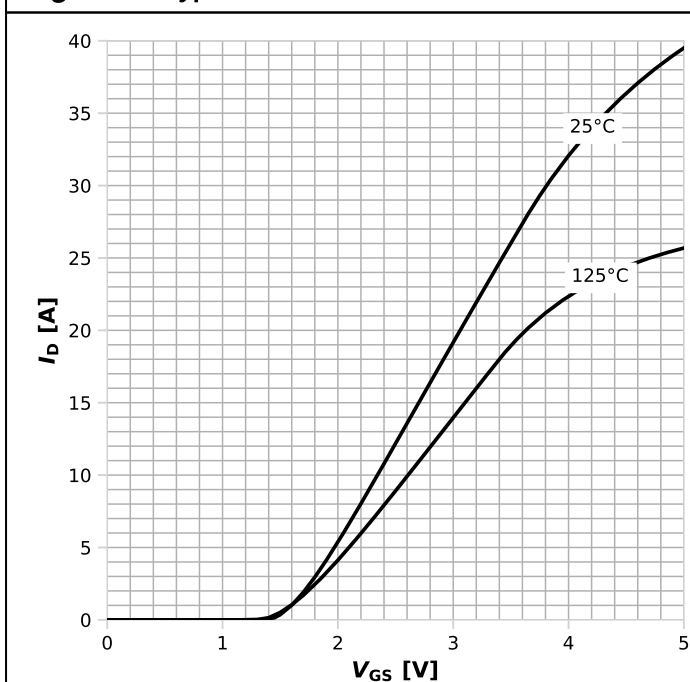
$I_{GS} = f(V_{GS}); \text{open drain}; \text{parameter: } T_j$

Diagram 12: Typ. gate characteristics reverse



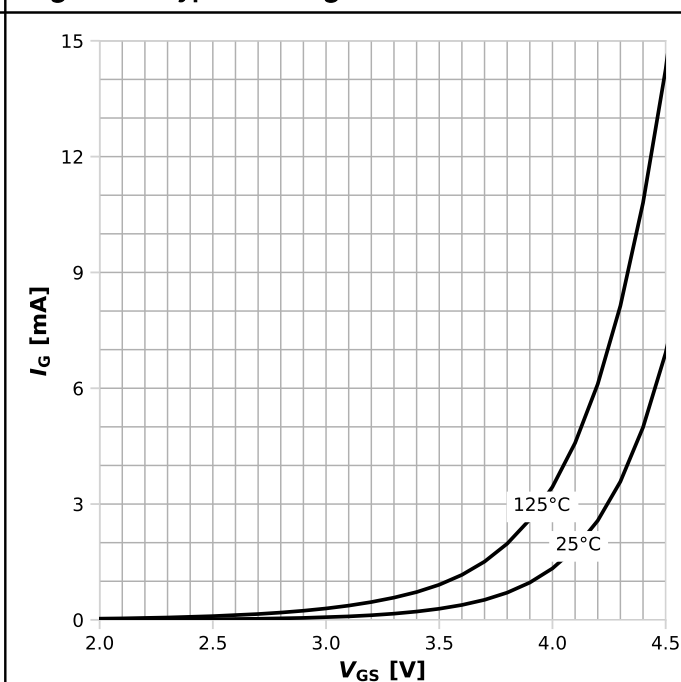
$I_{SG} = f(V_{SG}); \text{parameter: } T_j$

Diagram 13: Typ. transfer characteristics



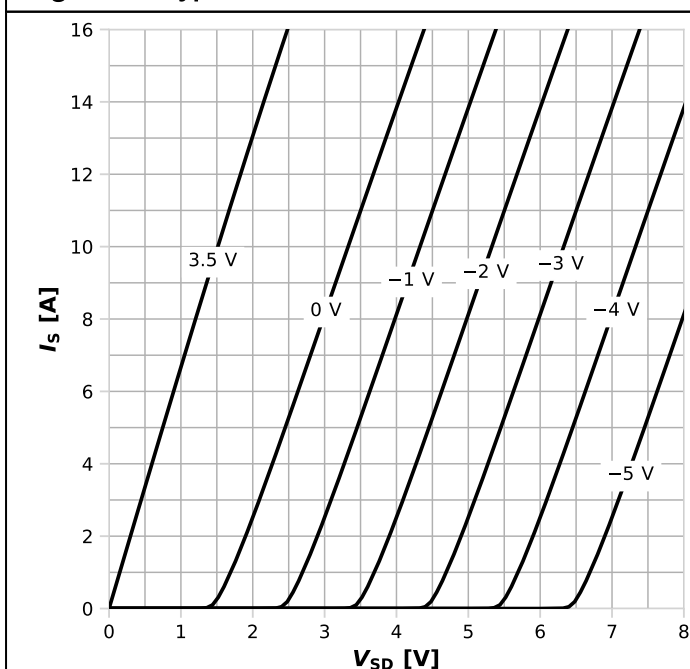
$I_D = f(V_{GS}); V_{DS} = 8V; \text{parameter: } T_j$

Diagram 14: Typ. transfer gate current characteristic



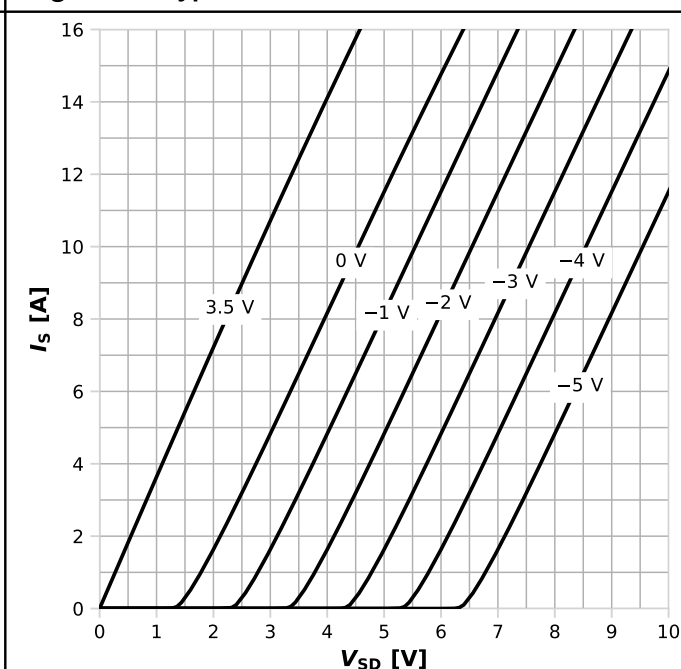
$I_G = f(V_{GS}); V_{DS} = 8V; \text{parameter: } T_j$

Diagram 15: Typ. channel reverse characteristics



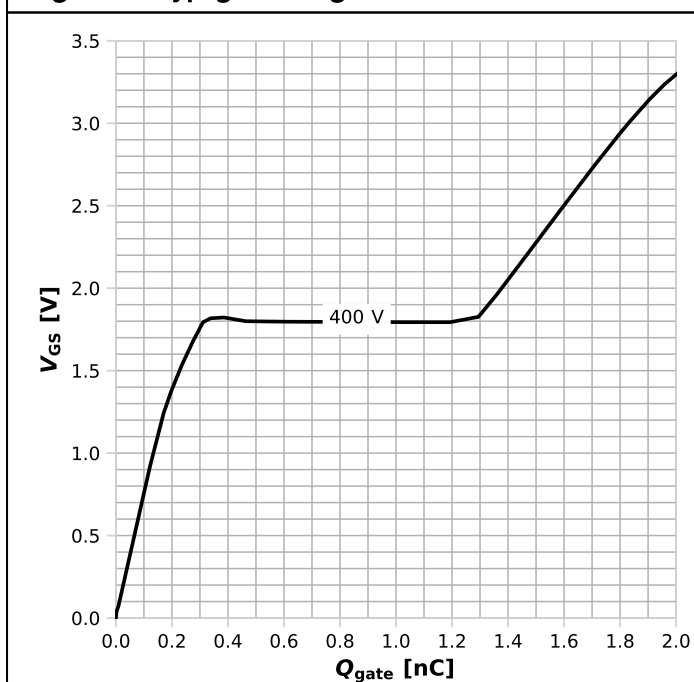
$I_S = f(V_{DS}); T_j = 25^\circ\text{C}; \text{parameter: } V_{GS}$

Diagram 16: Typ. channel reverse characteristics



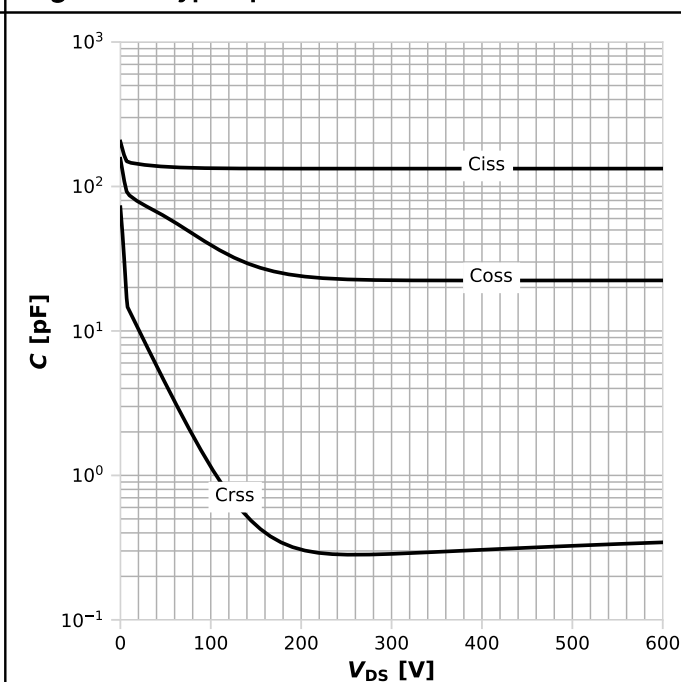
$I_S = f(V_{DS}); T_j = 125^\circ\text{C}; \text{parameter: } V_{GS}$

Diagram 17 Typ. gate charge



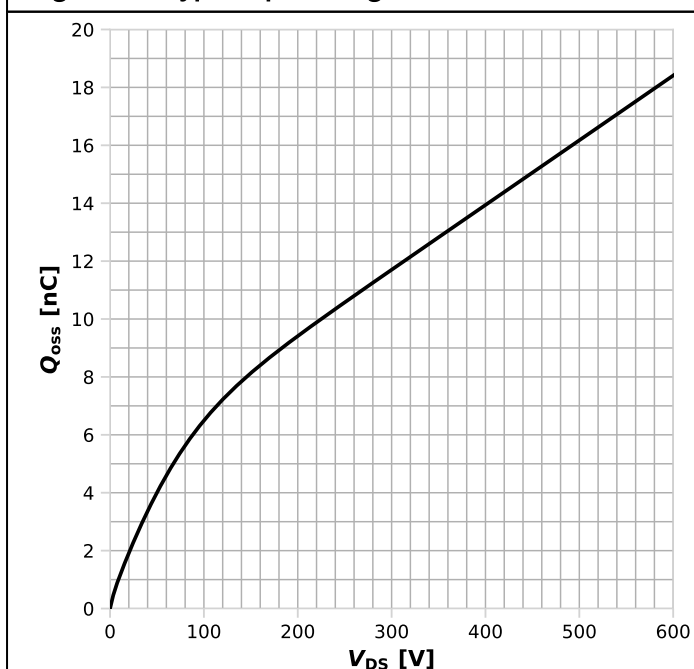
$V_{GS}=f(Q_{gate})$; $I_D=3.1$ A pulsed; $I_G=4.5$ mA; parameter: V_{DD}

Diagram 18: Typ. capacitances



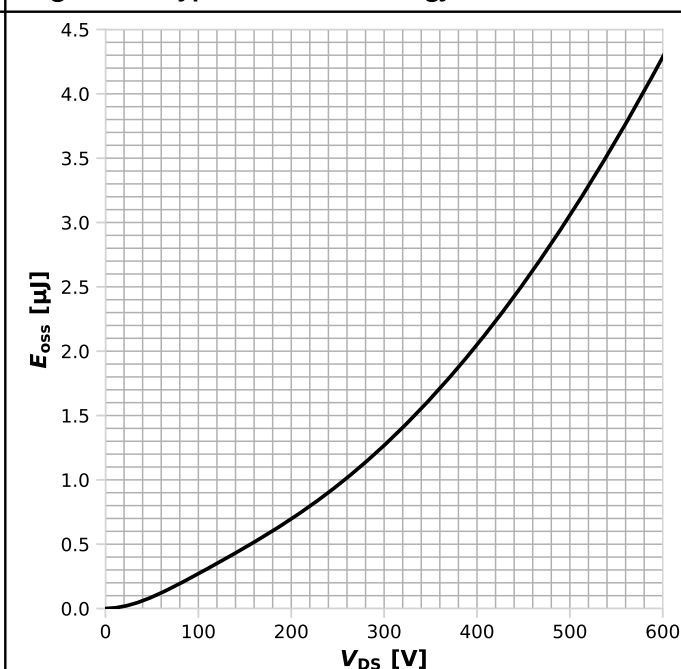
$C=f(V_{DS})$; $V_{GS}=0$ V

Diagram 19: Typ. output charge



$Q_{oss}=f(V_{DS})$

Diagram 20: Typ. Coss stored energy



$E_{oss}=f(V_{DS})$

5 Test circuits

Table 8 Reverse channel characteristics test

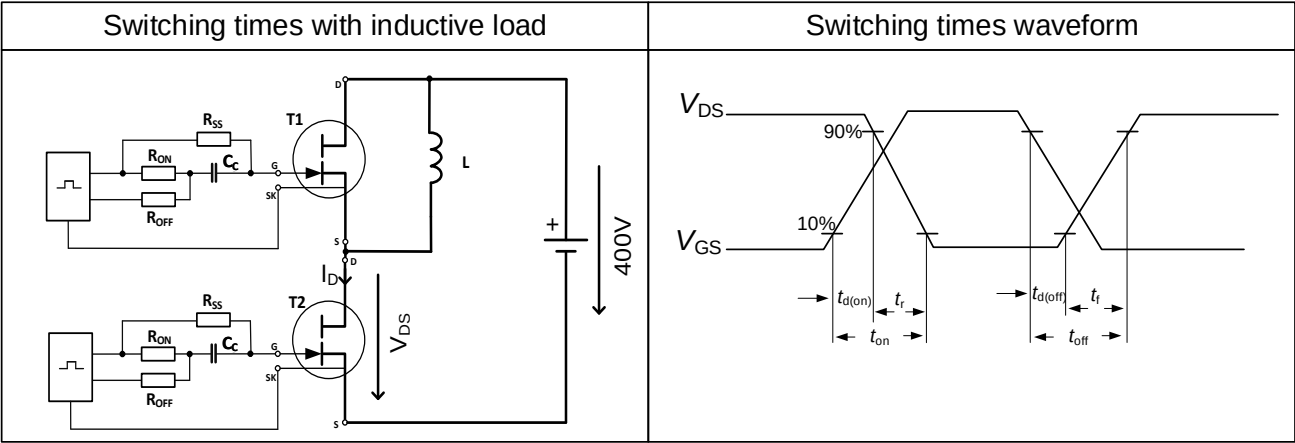
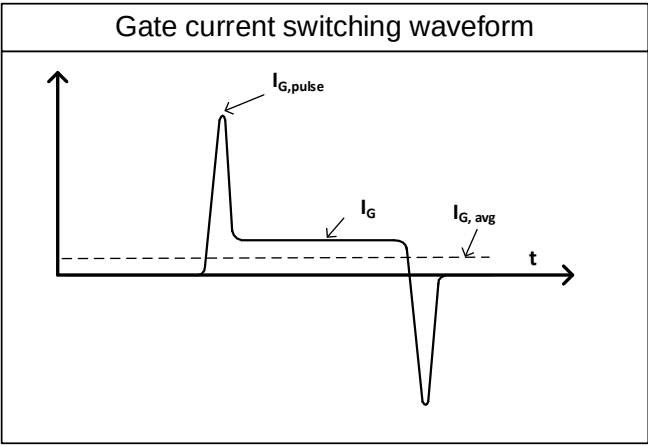
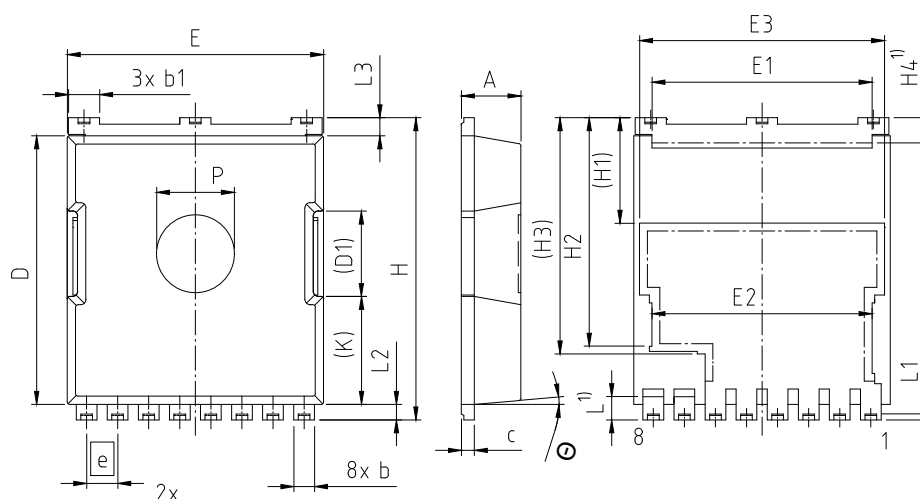


Table 9 Gate current switching waveform



6 Package outlines

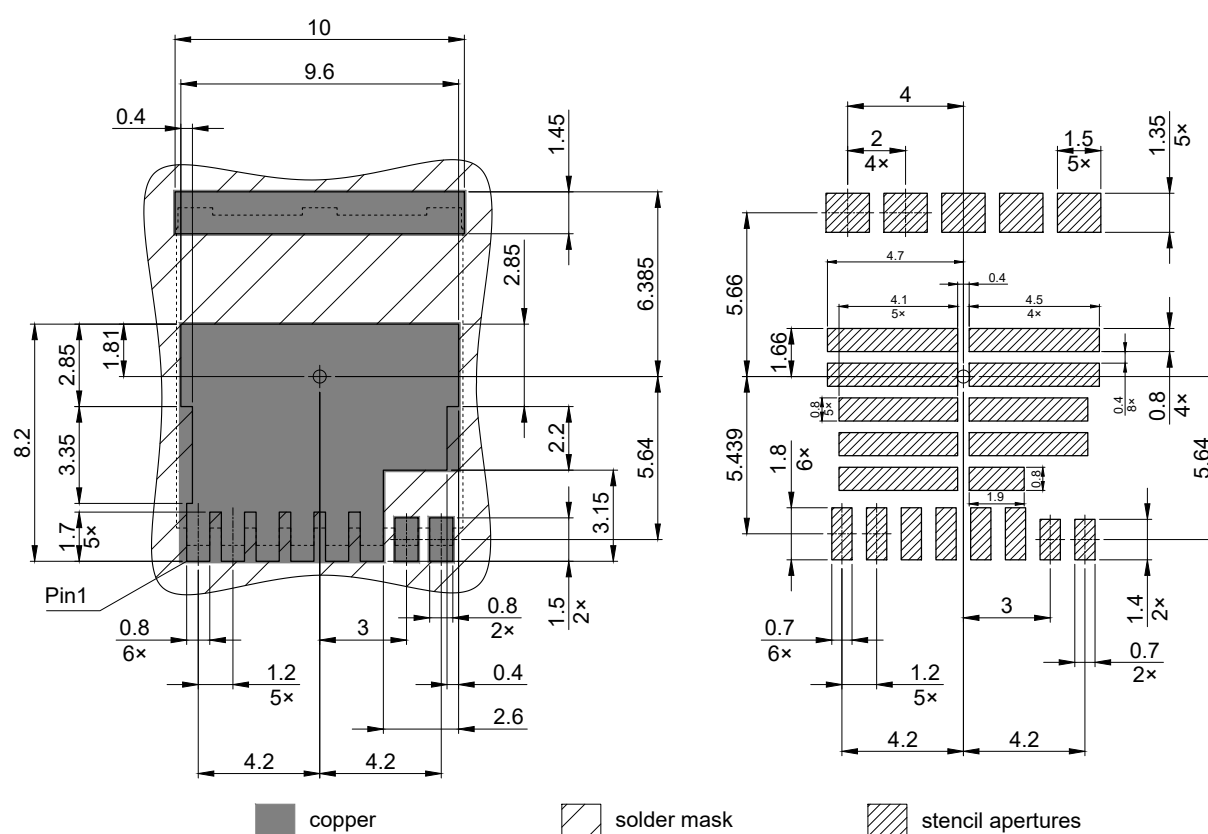


PACKAGE - GROUP NUMBER: PG-HSOF-8-U04		
DIMENSIONS	MILLIMETERS	
	MIN.	MAX.
A	2.20	2.40
b	0.70	0.90
b1	1.10	1.30
c	0.40	0.60
D	10.275	10.575
D1	(3.20)	(3.40)
E	9.70	10.10
E1	8.40	8.60
E2	8.40	8.60
E3	9.36	9.56
e	1.20	
H	11.475	11.875
H1	(3.98)	(4.18)
H2	8.73	8.93
H3	(9.03)	(9.23)
H4	0.88	1.08
N	8	
K	(4.07)	(4.27)
L	0.80	1.00
L1	0.13	0.33
L2	0.50	0.70
L3	0.60	0.80
P	2.90	3.10
Θ	3.5°	6.5°

NOTES:

1) LEAD LENGTH UP TO ANTI FLASH PROFILE, MOLD FLASHES EXCLUDED

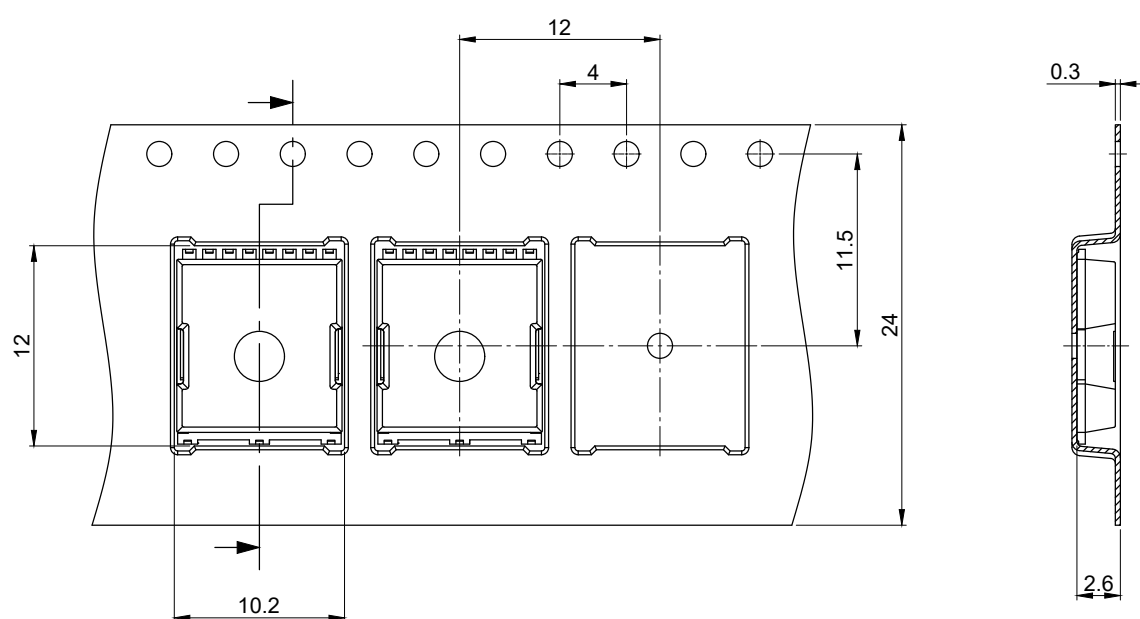
Figure 1 Outline PG-HSOF-8, dimensions in mm



Based on stencil thickness 0.130 mm

All dimensions are in units mm

Figure 2 Footprint drawing PG-HSOF-8, dimensions in mm



All dimensions are in units mm

The drawing is in compliance with ISO 128-30, Projection Method 1 [—

Figure 3 Packaging variant PG-HSOF-8, dimensions in mm

7 Appendix A

Table 10 **Related links**

- CoolGaN™ webpage
- CoolGaN™ reliability white paper
- CoolGaN™ gate driver application note
- CoolGaN™ applications information
- Package information

Revision history

IGT65R140D2

Revision 2026-03-05, Rev. 1.1

Previous revisions

Revision	Date	Subjects (major changes since last revision)
1.0	2024-11-28	Release of final
1.1	2026-03-05	Update of format and footnotes. Typical gate current added.

Trademarks

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